



Git & GitHub Cheat Sheet

+ Hands-On Exercises



1. Git Foundations

Concepts: Version control, Staging Area, Repo

Key Commands:

- `git init`
- `git add <file>`
- `git commit -m "msg"`
- `git log --oneline`

Exercise: Init repo, create `README.md`, commit.

Checkpoint: Explain staging area vs. repo.

3. Remote Collaboration

Concepts: Clone, Fetch, Pull, Push

Key Commands:

- `git clone <repo>`
- `git pull origin main`
- `git push origin <branch>`

Exercise: Clone repo, push branch, open PR.

Checkpoint: Fetch vs. Pull?

5. Enterprise & CI/CD

Concepts: GitHub Flow, CI Pipelines

Sample YAML:

```
name: CI
on: [push, pull, request]
jobs:
  build:
    - uses-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v3
    - run: ./deploy.sh
```

Exercise: Set up CI, merge after tests pass.

Checkpoint: What event triggers CI?

2. Branching & Workflow

Concepts: Branches, Merge, Rebase, Stash

Key Commands:

- `git checkout -b <branch>`
- `git merge <branch>`
- `git rebase -i HEAD~3`
- `git stash`

Exercise: Create 'feature/login', merge to 'main' & Resolve conflict

Checkpoint: Merge vs. Rebase?

4. Advanced Git

Concepts: Reflog, Reset, Aliases

Key Commands:

- `git reflog`
- `git reset --soft HEAD~1`
- `git config alias.co checkout`

Exercise: Undo last commit, recover via reflog.

Checkpoint: Reset vs. Revert?

6. Productivity Tips

Commands:

- `git diff main feature/login`
- `git stash push -m "WIP"`
- `git log --graph -all`

Exercise: Stash changes, compare branches.

Checkpoint: Purpose of stashing?

References:

- Pro Git Book: git-scm.com/book
- GitHub Flow: docs.github.com
- GitHub Actions: docs.github.com/actions